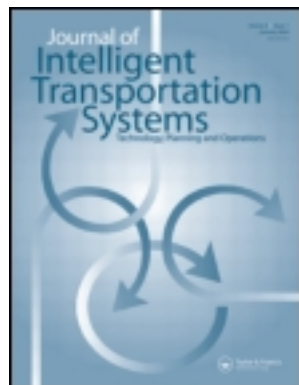


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Study on Eco-Route Planning Algorithm and Environmental Impact Assessment

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The transportation sector has faced increased environmental pressure due to the rapid motorization in China over recent years. In addition to energy policies and the use of fuel-efficient vehicles, CO₂ emissions and fuel consumption can be lowered through the application of intelligent transportation systems (ITS) technologies that can affect drivers' route choice decision-making process and reduce congestion. Ecological route (eco-route) planning is one such strategy that provides the least CO₂ emission and fuel consumption routes for vehicles. In this article, a set of mesoscopic vehicle emission and fuel consumption models are established based on locally collected vehicle operation and emission data, and these models are more appropriate for evaluating the effectiveness of ITS strategies with sufficient estimation accuracy. Based on the proposed models and a dynamic traffic information database, an eco-route planning algorithm is proposed as well, which is expected to be consistent with the road network characteristics of China cities. Furthermore, the environmental impact of the eco-route is fully evaluated with tests undertaken in Beijing urban areas. The results show that the proposed eco-route algorithm significantly reduces fuel consumption and has good environmental performance.

Keywords Eco-Route Planning Algorithm; Environmental Impact Assessment; Real-Time Traffic Information; Vehicle Emission Model

INTRODUCTION

With rapid urbanization and motorization, the amount of vehicle kilometers traveled (VKT) continues to increase across China recently. The transportation sector consumes a large amount of energy and is responsible for approximately 9% of the nationwide consumption in 2008 (Wang, 2010). Given the dual pressure from increased energy demand and sustainable development mandate, it is inevitable for China to develop a low-carbon transportation system, which is characterized by low energy consumption, low pollution, and low emission.

For low-carbon transportation solutions, some well-known methods such as establishing strict energy policies, developing public transport systems and promoting electric vehicles have

been studied and adopted widely. Recently, with the development of intelligent transportation systems (ITS) technologies, increased attention has been paid to the indirect effect of ITS technologies on reducing fuel consumption and CO₂ emissions, besides the optimal assignment of traffic flow or improvement of road traffic conditions. As one of the well-applied ITS technologies, a navigation system can provide the route guidance with minimum travel distance, minimum travel time, or the least generalized cost that incorporates both distance and the number of routing options (Zhu, Boriboonsomsin, & Barth, 2010), but the results of route planning are not always optimal from an environmental and energy perspective. This article attempts to establish an eco-route (ecological route) planning algorithm that is designed particularly for minimizing vehicle emissions and fuel consumption.

This article is organized as follows: The second section summarizes the previous work on emission modeling and Eco-route planning. In the third section, based on the emission data collected by a portable emission measurement system (PEMS), a series of mesoscopic vehicle emission (i.e., carbon dioxide

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[CO₂], carbon monoxide [CO], hydrocarbons [HC], and oxides of nitrogen [NO_x] and fuel consumption models is established. The fourth section proposes the algorithm of eco-route planning. Based on the data collected in Beijing central city, the environmental performance of the proposed eco-route algorithm is evaluated in the fifth section, and the conclusions are given in the final section.

LITERATURE REVIEW

Recent literature shows that the faster highway route choice is not always the optimal route from an environmental and energy consumption perspective (Ahn & Rakha, 2007). Selecting different travel routes between the same origin and destination can result in different vehicle emissions and fuel consumption (Ahn & Rakha, 2008). An exploratory study in Sweden finds that the driver's spontaneous route choice is not the most fuel-efficient for 46% of trips in Lund. These trips could have saved 8.2% fuel by using a fuel-optimized navigation system (Ericsson, Larsson, & Brundell-Freij, 2006).

Transport researchers have sought to explore the impact of route choice changes on reducing vehicle emissions and fuel consumption, and to find the most environmentally friendly routes. There are many factors affecting vehicle emissions and fuel consumption, such as vehicle characteristics, road characteristics, and traffic conditions. Total vehicle emissions and fuel consumption during a trip are sensitive to changes in travel speed and operating mode distribution. Changes in route choices for different road types and various traffic conditions may lead to significant variation of environmental costs. Ahn and Rakha (2007) demonstrate that a small portion of the entire trip that involves high engine-load conditions has significant impacts on the total emissions, and it is expected that the route planning method can search for an eco-route with the smallest emissions and fuel consumption. Ahn and Rakha (2008) investigate the impacts of route choice decisions on vehicle energy consumption and emission rates using microscopic and macroscopic emission estimation tools for a simple network. They also find that macroscopic emission estimation tools (e.g., MOBILE6) can lead to erroneous conclusions.

Kono et al. (2008) propose an ecological route searching model that generates routes with the least fuel consumption, in which many factors such as traffic information, geographic information, and vehicle parameters are comprehensively considered. In this model, the impacts of vehicles' movement as well as factors affecting basic fuel consumption are used to express the energy loss. Some methods (Barth, Boriboonsomsin, & Vu, 2007) that incorporate the Comprehensive Modal Emission Model (CMEM) with route minimization algorithms are applied in several case studies in southern California, and significant energy savings compared to the standard navigation techniques are observed. Boriboonsomsin et al. (2010) develop an eco-routing navigation system that includes a dynamic road network database, an energy/emissions operational parameter set, a routing engine, and user interfaces. In addition, some

advanced navigational tools have been emerging on the Internet. For example, Navitime Corporation in Japan has developed a trial version for searching out an "eco-route" that is available on the Internet to the public.

Link cost function is the most essential factor for shortest-path planning algorithms. For eco-route planning, the link cost function reflects the value of emissions or fuel consumption, and the influence of the vehicle speed, acceleration, and other driving conditions should be taken into account. In accordance with driving parameters, the models for predicting emissions can be divided into two types, microscopic models and mesoscopic models. Microscopic driving parameters reflect the driving characteristics of vehicles in the context of each point of time, such as instantaneous speed and instantaneous acceleration. Mesoscopic traffic parameters reflect the driving characteristics of vehicles in the context of a period of time, such as average speed and proportion of acceleration or deceleration. The U.S. federal government's MOBILE6 and California's EMFAC (2007) estimate emission factors based on vehicle driving schedules and link speeds mesoscopically. These models are intended to predict emission inventories for large regional areas, but they are not well suited for evaluating traffic operational improvements that in nature are more "microscopic" (Barth et al., 2001). Thus, these models lack considerations of the impact of vehicles' operating mode on emissions, which makes these models unable to reflect the characteristics of vehicle emissions under different traffic conditions. The CMEM model and VT-Micro (Ahn et al., 1999) model are microscopic tools for estimating the emissions accurately. CMEM utilizes vehicle operational variables (such as speed, acceleration, and road grade) and model-calibrated parameters (such as cold-start coefficients and engine-friction factor) as input data (Barth et al., 2000). However, it is difficult to obtain such information in reality. Also VT-Micro requires substantial amounts of input data such as second-by-second vehicle velocity profiles. As a result, this type of model may not be applicable for dynamical route planning. Therefore, it is more appropriate to utilize link-based emission models with consideration of transient vehicle behavior to improve both the applicability and accuracy of the evaluation effect of local ITS strategies.

Many studies have found more accurate approaches to estimate vehicle emissions and fuel consumption. In order to characterize the relationship between fuel consumption/vehicle emissions and vehicle driving parameters, methods based on vehicle specific power (VSP) have been proposed. MOVES is the most recent emission model developed by the U.S. Environmental Protection Agency (EPA) as an attempt to replace existing emission models including MOBILE (Koupal et al., 2002). It incorporates the concept of VSP bins and characterizes vehicle activities according to VSP and speed. Such a VSP-based emission model not only has a better estimation of vehicle emissions than a speed-based emission model but also is capable for reflecting the emission changes under different operating modes. Wang et al. (2008) establish a model based on VSP, which can be used to estimate the vehicle fuel consumption, given the driving parameters. It provides insight on the impact mechanism of

levels of cruise speed and acceleration on vehicle fuel consumption. Further, a speed and VSP modeling method is proposed to improve the accuracy with prediction errors within 20% for trip emissions and link-speed-based emission factors (Wang & Fu, 2010).

As demonstrated, link-based emission models are necessary for considering the utilization of traffic status parameters in most existing road traffic information systems. Meanwhile, the second-by-second emission rate, along with the dynamic traffic condition, also remains an important characteristic that needs to be considered in the models. Many studies have focused on vehicle emission and fuel consumption estimation models for eco-route planning. However, these models divide vehicle speed into different levels, and the speed categories are too broad to be integrated with the variation of actual operation modes. In addition, the input data of these models are generally average speed of a whole trip or a period of time such as one hour, and the time ranges are too wide to take full advantage of the probe vehicle technology or other traffic information systems. Moreover, parameters of emission and fuel estimation models should be established according to the characteristics of the road network and vehicle driving schedule in China. It is of significance to analyze the environmental impact assessment of different route choices and the effectiveness under various traffic conditions for a more comprehensive realistic network.

In order to overcome the limitations of current methods for evaluating the environmental impact of eco-route planning, this study establishes emission and fuel consumption models using average speed as explanatory variables. In order to exclude microscopic parameters, different VSP division methods on emission estimation are introduced. Therefore, real-time vehicle emissions can be estimated by binning instantaneous VSP in this study. The variance of vehicle emissions and fuel consumption to different average travel speeds is described successfully in these models. Furthermore, environmental impact assessment of the eco-route and the analysis of effectiveness are executed for a more comprehensive realistic network.

VEHICLE EMISSIONS FACTOR AND FUEL CONSUMPTION MODELS

Microscopic emission models are quite complex and detailed as they provide instantaneous vehicle emission rates, but they may not be practical for dynamic route guidance systems because the detailed vehicle traveling information (e.g., vehicles' speed, acceleration level, and road grade) is not available in most existing road surveillance systems. Compared to microscopic emission models, mesoscopic models estimate link-based emissions or fuel consumption, which are more suitable for ITS applications. However, most of the models based on average speed usually ignore the comprehensive effects from vehicles' speed, acceleration level, and road grade.

This article attempts to establish a series of mesoscopic models for light-duty gasoline vehicles, mid-duty diesel vehicles, and heavy-duty diesel vehicles separately based on the emis-

sion data collected by PEMS under actual driving conditions. The influence of the vehicle speed, acceleration, and other driving conditions on the vehicle emissions is given full consideration through analyzing the VSP distribution for each travel speed level. Final vehicle emission factor models and fuel consumption models are expected to be described as functions of average link speed. The following subsections describe how these models are established.

Methodology

VSP is defined as the instantaneous power per unit mass of the vehicle and it is considered as a core of MOVES (Koupal et al., 2002). The equation is as follows (Jiménez-Palacios, 1999):

$$VSP = v \times (a(1 + \varepsilon_i) + g \times \text{grade} + g \times C_R) + \frac{1}{2} \rho_a \frac{C_D \times A}{m} (v + v_w)^2 v \quad (1)$$

where v is the vehicle speed, m/s; a is the vehicle acceleration, m/s²; and ε_i is the "mass factor," which is the equivalent translational mass of the rotating components (wheels, gears, shafts, etc.) of the powertrain. The suffix i indicates that ε_i is gear dependent; g is the acceleration of gravity, 9.81 m/s²; grade is the road grade; C_R is the coefficient of rolling resistance; ρ_a is the ambient air density, 1.207 kg/m³ at 20°C; C_D is the drag coefficient; A is the frontal area of the vehicle, m²; m is the vehicle mass, kg; and v_w is the headwind into the vehicle, m/s.

The simplified function (Jiménez-Palacios, 1999) in Eq. (2) is used for the VSP calculations of light-duty gasoline vehicles and mid-duty diesel vehicles. Referring to the previous study (Andrei, 2001), Eq. (3) is used for heavy-duty diesel vehicles' VSP calculation:

$$VSP = v \times (1.1 \times a + 0.132) + 0.000302 \times v^3 \quad (2)$$

$$VSP = v \times (a + 0.09199) + 0.000169 \times v^3 \quad (3)$$

where v and a are the vehicle speed and acceleration in m/s and m/s², respectively.

Some current models estimate vehicle emissions and fuel consumption based on the data collected by a chassis dynamometer test according to typical urban driving cycles such as ECE + EUDC. However the typical driving cycles are not consistent with the real driving situation on urban roads in China. We use the data collected by vehicles with PEMS during peak and nonpeak hours of city traffic in this study. The driving conditions are different, with speed ranging from 0 to 100 km/hr and acceleration from -5 m/s² to 5 m/s², while the corresponding VSP calculated using second-by-second data is between -40 kw/t and 40 kw/t.

For the characteristics of vehicle emission change under different travel conditions, VSP is separated as a bin. Each VSP bin is associated with an average emission rate. The second-by-second data collected by PEMS is divided into traveling fragments by 60-s intervals and each fragment is characterized with

its average speed. After the interval division of average speed, the VSP-Bin distribution attribute of each average speed interval is calculated. The average emission rate under each average speed bin is estimated as:

$$\overline{ER}_i = \sum_j ER_j \times D_{ij} \quad (4)$$

where $\overline{ER}_i$ is the average emission rate for average speed bin i , g/s; j is the VSP bin index; ER_j is the emission rate for VSP bin j , g/s; and D_{ij} is the VSP-Bin distribution for average speed bin i and VSP bin j .

Then the emission factor under each average speed bin is estimated as:

$$V_l = \sum_{k=1}^n D_k / \sum_{k=1}^n T_k \quad (5)$$

$$EF_l = ER_i \times \sum_{k=1}^n T_k / \sum_{k=1}^n D_k \quad (6)$$

where T_k is the vehicle trip time spent in travel fragment k of average speed bin i , s; D_k is the vehicle trip distance in travel fragment k of average speed bin i , km; V_l is the vehicle travel speed for average speed bin i , km/hr; ER_i is the emission rate for average speed bin i , g/s; and EF_l is the emission factor for V_l .

Vehicle mesoscopic emission factor models for each vehicle type are estimated using the multiple linear regression approach,

and an emission factor is represented by a function of average speed (Namikawa, Takai, & Ohshiro, 2003):

$$EF = \frac{a}{v} + bv + cv^2 + d \quad (7)$$

where EF is the emission factor, g/km; v is average speed, km/hr; and a , b , c , and d are coefficients.

RESULTS

The emission and fuel consumption models are established by using the PEMS data based on the proposed approach of vehicle mesoscopic emission factor models. The four types of vehicle emission curves for each vehicle type are shown in Figure 1.

Figure 1 shows that the shapes for all vehicle emission curves are consistent with the exiting research results (Ahn et al., 1999; André & Rapone, 2009; Zegeye et al., 2009): Vehicle emission factors decrease as speed increases to a specified value, and then start to increase. For example, for the light-duty vehicle, the value of CO₂ emission per kilometer declines greatly with the increase of speed. Once the speed reaches to about 60 km/hr, the value of CO₂ emission per kilometer will increase slowly. However, both higher and lower speed will lead to a higher vehicle emission rate. Approximately, the average speed of about 60 km/hr results in a minimum CO₂ emission rate of 110 g/km. Moreover, the optimal vehicle speed turns out to be around

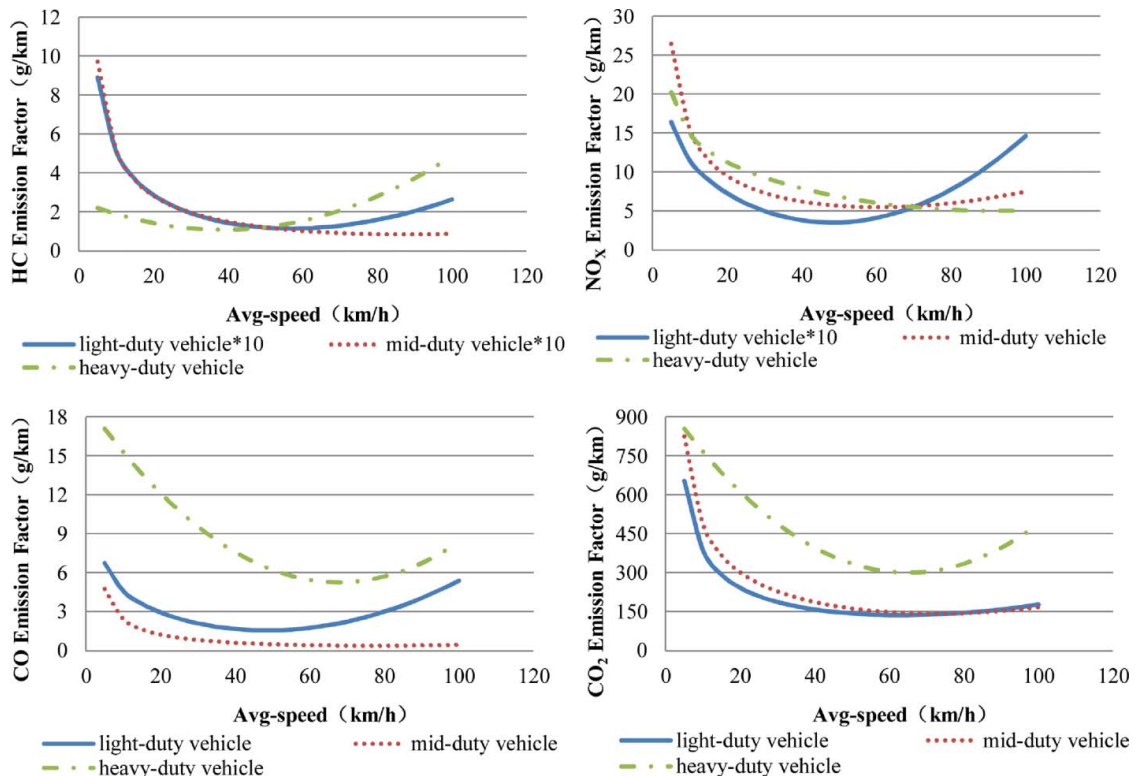


Figure 1 Vehicle emission curves of different vehicle types (color figure available online).

Table 1 Vehicle emission factor models.

Emission	Vehicle type	V^2 (t -value)	V (t -value)	V^{-1} (t -value)	Constant (t -value)	Adjusted R^2
HC	Light-duty	1.566×10^{-5} (5.144)	-1.223×10^{-3} (-3.336)	1.581 (32.833)	2.721×10^{-2} (2.566)	0.99
	Mid-duty	1.483×10^{-5} (3.586)	-1.880×10^{-3} (-6.587)	4.432 (328.330)	9.580×10^{-2} (19.898)	0.99
	Heavy-duty	6.617×10^{-4} (6.066)	-7.741×10^{-2} (-8.350)	—	2.572 (15.353)	0.88
NO _x	Light-duty	1.765×10^{-5} (4.144)	-1.861×10^{-3} (10.536)	1.019 (-3.528)	5.113×10^{-2} (3.148)	0.96
	Mid-duty	9.235×10^{-2} (2.788)	-9.235×10^{-2} (-5.449)	1.063×10^2 (132.606)	5.653 (19.831)	0.99
	Heavy-duty	9.472×10^{-4} (2.425)	-1.687×10^{-1} (-4.197)	4.540×10 (6.903)	1.202×10 (11.046)	0.98
CO	Light-duty	1.155×10^{-4} (5.853)	-9.170×10^{-3} (-3.860)	4.866 (15.593)	2.558×10^{-1} (3.721)	0.97
	Mid-duty	7.102×10^{-5} (3.893)	-7.102×10^{-5} (-5.512)	2.291×10 (384.815)	2.103×10 (9.935)	0.99
	Heavy-duty	3.063×10^3 (6.672)	-4.070×10 (-10.431)	—	1.908×10 (27.063)	0.94
CO ₂	Light-duty	2.949×10^{-2} (2.233)	-3.061 (-1.925)	3.952×10^3 (18.918)	2.233×10^2 (4.853)	0.98
	Mid-duty	2.681×10^{-2} (2.838)	-3.294 (-5.053)	3.237×10^2 (104.952)	1.948×10^2 (17.767)	0.99
	Heavy-duty	1.540×10^{-1} (5.675)	-2.002×10 (-8.684)	—	9.514×10^2 (22.833)	0.92

Note: EF is the emission factor, g/km; V is average speed, km/h.

50 km/hr (55 km/hr, 55 km/hr) for a minimum CO (NO_x, HC) emission rate of 1.56 g/km (0.36 g/km, 0.12 g/km). Comparing our models and results with the study of Zegeye et al. (2009), the latter has a higher optimal vehicle speed of 65 km/hr (55 km/hr, 85 km/hr) for a lower minimum CO (NO_x, HC) emission rate of 1.0 g/km (0.32 g/km, 0.09 g/km).

The coefficients and t -values of the variables (V^2 , V , and V^{-1}) for specific formulas are shown in Table 1.

For the three types of vehicles, the curves indicate a second- or third-order polynomial model structure between emission rates and average speed. The statistical results indicate a good fit for emission estimation as demonstrated in Table 1 (R -squared values in excess of .88 for all emissions of different types of vehicles). Based on the carbon balance method listed in national standards of China (GB/T 19233-2003), vehicle fuel consumption factors for gasoline vehicles and diesel vehicles can be estimated using Eqs. (8) and (9), respectively:

$$FC_S = 0.1154 [(0.866 \times EF_{HC}) + (0.429 \times EF_{CO}) + (0.273 \times EF_{CO_2})] \quad (8)$$

$$FC_C = 0.1155 [(0.866 \times EF_{HC}) + (0.429 \times EF_{CO}) + (0.273 \times EF_{CO_2})] \quad (9)$$

where FC_S is the fuel consumption factor of gasoline vehicles, kg/100 km; FC_C is the fuel consumption factor of diesel vehicles, kg/100 km; EF_{HC} is the HC emission factor, g/km; EF_{CO} is the CO emission factor, g/km; and EF_{CO_2} is the CO₂ mission factor, g/km.

The vehicle fuel consumption factor models are summarized in Table 2.

ECO-ROUTE PLANNING ALGORITHM

A typical route planning algorithm searches for the route with the shortest time or shortest distance based on a specific optimization goal. However, considering the relationship between the emissions and vehicle speed listed in Figure 1, these route

planning algorithms do not always guarantee the routes with the least emissions and fuel consumption. Though there are a number of studies about methods of searching for an eco-route, most of them ignore the impact of dynamic changes of traffic conditions on route planning and lack application of the eco-route planning to a more comprehensive realistic network. In this study, based on the classic Dijkstra's shortest path algorithm, an improved dynamic route searching algorithm is utilized to explore the most environmentally friendly route. The proposed route planning algorithm is characterized with its dynamic estimation of emission and fuel consumption of each link, by using dynamic traffic information updated every 5 min. Furthermore, the proposed algorithm can be utilized to assess the environmental impact of different route choices (i.e., eco-route or shortest-time route) for more comprehensive realistic networks.

Dynamic Traffic Information Database

For the purpose of accurately estimating link costs, we establish a dynamic traffic information database to reflect the temporal variation of traffic information. In this study, topology information of road networks referring to the static characteristics (e.g., linkage relationship between links, link length, road grade, direction, etc.) is linked with the real-time road traffic information. The real-time road traffic information (i.e., link-based average speed in five-minute intervals) is collected from a probe vehicle system, which provides more accurate estimates of travel time for each link (Lin, He, & Kornhauser, 2008). Furthermore, the link travel time is updated every 5 min dynamically. The dynamic traffic information used for vehicle emission

Table 2 Vehicle fuel consumption factor models.

Fuel consumption	Light-duty vehicle	$FC_S = 125.015/V - 0.097V + (9.220 \times 10^{-4})V^2 + 7.056$
	Mid-duty vehicle	$FC_C = 103.633/V - 0.104V + (8.560 \times 10^{-4})V^2 + 6.163$
	Heavy-duty vehicle	$FC_C = -0.659V + 0.005V^2 + 31.201$

Note. FC is the fuel consumption factor, kg/100 km; V is average speed, km/hr.

or fuel consumption estimation is predicted based on a history database combined with real-time road traffic information, and is stored in the dynamic traffic information database.

Link Cost Based on Emissions and Fuel Consumption

In the eco-route planning process, link-based emissions or fuel consumption is used as link cost, so assigning specific emission factors and fuel consumption factors for each link in road networks plays a key role. According to the mesoscopic emission and fuel consumption models established in the preceding section, the emission and fuel consumption factors are described as functions of average link speed. Therefore, link cost for eco-route planning can be calculated based on the dynamic traffic information of the road networks and link length.

For example, the vehicle emissions of CO₂ and fuel consumption for link ln can be calculated by Eqs. (10) and (11), respectively:

$$C_{vc,t,ln}^{EF} = EF_{CO_2} \times l_{ln} \quad (10)$$

$$C_{vc,t,ln}^{FC} = FC \times l_{ln} \quad (11)$$

where vc is the vehicle type; t is the current time window; EF is the emission factor, g/km; FC is the fuel consumption factor, kg/100 km; and l is the length of link ln .

Finally, based on the link cost matrix for shortest-time path planning (i.e., the dynamic traffic information database), in which the travel time for each link of the road network at each 5-min interval for a whole day (i.e., 288 intervals of 5 min) is saved in a cell, the link cost matrix for eco-route planning can also be built to illustrate the dynamic emissions or fuel consumption for all links in road networks at each 5-min interval.

Route Planning Algorithm Based on Improved Dijkstra

For any specific origin, destination, and departure time, a shortest-time path or an eco-route can be calculated based on the travel time or emission (fuel consumption) link cost matrix using Dijkstra's algorithm. In order to make the algorithm more efficient and practical, some techniques such as the least heap structure are utilized. The result indicates that a significant reduction in calculating time is achieved with the improved algorithm. During the process of route searching, the adequate cell in link cost matrix is found according to the entering time and the link number, and then the values of travel time, emissions or fuel consumption in the cell are acquired and used as the link cost of route planning.

Finally, a shortest-time path or an eco-route can be searched. The entire travel time, travel distance, vehicle emissions, and fuel consumption of the two routes are estimated simultaneously.

The total fuel consumption for a route can be calculated by Eq. (12):

$$T = \sum_{ln=1}^n C_{vc,t,ln}^{FC} \quad (12)$$

where T is the total fuel consumption for a route; n is the number of links included in the route; and $C_{vc,t,ln}^{FC}$ is the fuel consumption for link ln .

CASE STUDIES AND RESULTS

Considering that the shortest-time or shortest-distance routes do not always guarantee the least vehicle emissions and fuel consumption, this study finishes with a case study. The purpose is to evaluate the proposed eco-route planning algorithm and its effectiveness on vehicle emission and fuel consumption reduction from an environmental perspective.

The study area, a part of the central area of Beijing, is surrounded by the red rectangle with each side about 10 km length in Figure 2, and the total area is about 100 square kilometers. The dynamic travel time information is collected by the probe vehicle system, which covers most of the roads in Beijing, and is saved in the cost matrix for shortest-time path planning. The link cost matrix for eco-route planning is also built with the method already illustrated.

Three types of routes are searched based on the following three optimization goals: (a) the least CO₂ emissions, (b) the lowest fuel consumption, and (c) the shortest time. Then the CO₂ emissions, fuel consumption, travel time, and travel distance are compared among these routes.

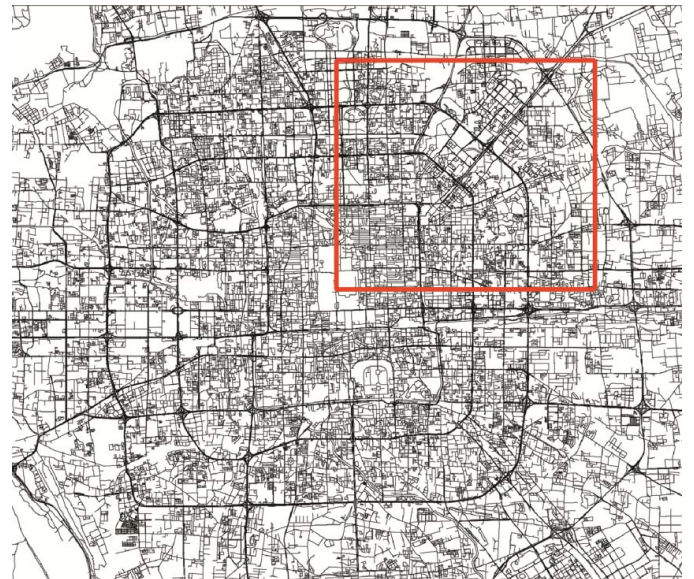


Figure 2 Studying area in Beijing city (color figure available online).

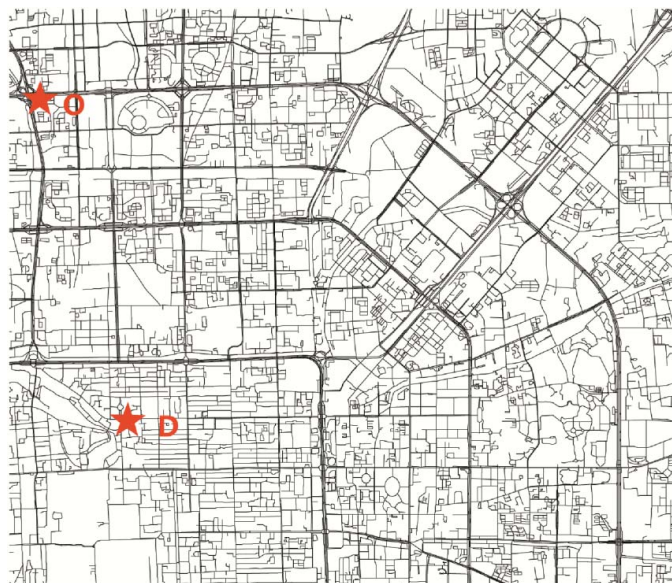


Figure 3 Origin and destination (color figure available online).

Comparison Between Shortest-Time Route and Eco-Route

In order to analyse the possible different effects of road traffic status on route planning, for a randomly selected origin-destination pair (OD pair) (shown in Figure 3), routes for light-duty vehicles with CO₂ emissions and travel time as the link costs are searched respectively with different departure times in a day. The values of vehicle emissions, fuel consumption, travel time, and distance are compared between the eco-routes

(i.e., least vehicle CO₂ emissions) and shortest-time routes (Figure 4).

There is no obvious difference in travel time for the eco-route and shortest-time route during all time periods. The result is consistent with the fact that the traffic status for all roads in the peak hours in Beijing is congested, and there is no great difference in travel time for all possible routes. However, obvious differences in CO₂ emissions, travel distance, and fuel consumption between the two routes can be observed in peak hours. For example, the eco-route has significant advantage in reducing CO₂ emissions and fuel consumption compared with the shortest-time route. There is no great difference in travel time, vehicle emission factors, and fuel consumption factors for all possible routes in peak hours, so the relatively obvious difference in CO₂ emissions and fuel consumption between the two routes mainly results from the difference in travel distance between the two routes.

Figure 5 illustrates the route planning results of the eco-route (green line) and the shortest-time route (blue line) for the same OD pair shown in Figure 3 in the morning peak hour period. The comparison results between the two routes are summarized in Table 3. Values in the table are normalized to the shortest-time route's results. The fuel consumption of the eco-route is about 6% lower than that of the shortest-time route, although the travel time of the eco-route is about 4% longer. This result verifies that the eco-route exactly reduces the CO₂ emissions.

Environmental Effectiveness Analysis

As described earlier, the amount of the emissions and the fuel consumption are affected by travel distance, emission factors,

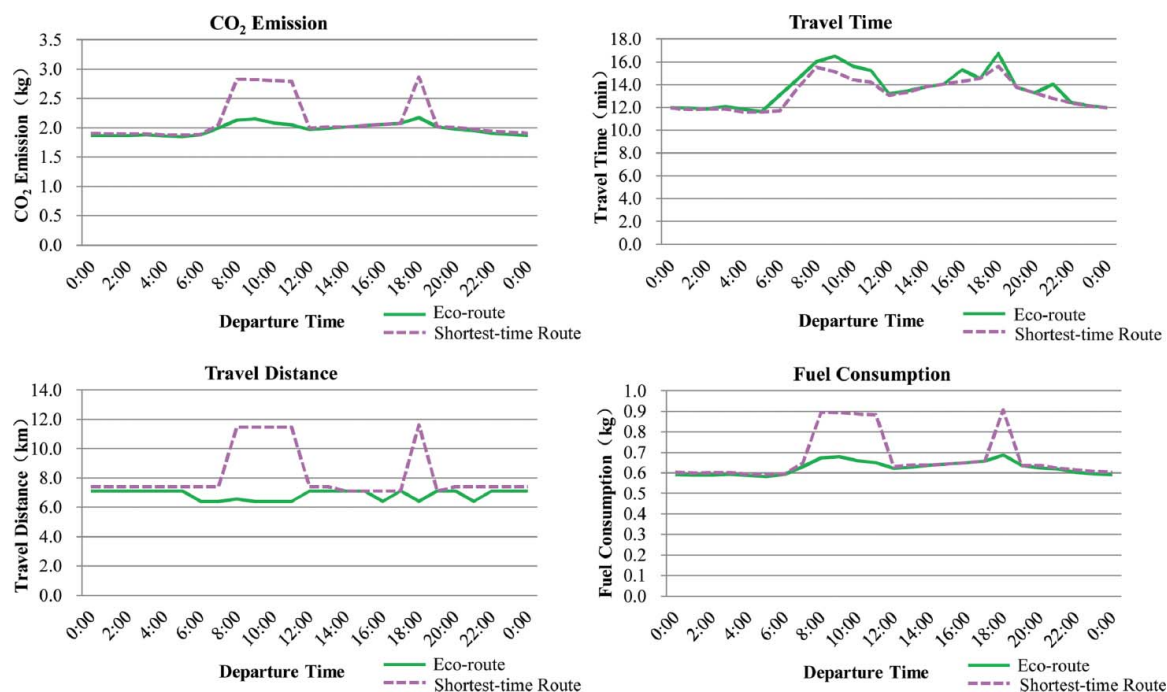


Figure 4 Comparison between eco-route and shortest-time route for different departure times (color figure available online).



Figure 5 Shortest-time route and eco-route for specific OD (color figure available online).

and fuel consumption factors, while the emission and fuel consumption factors are described as functions of travel speed. In order to analyze the environmental effectiveness under different travel distance and average speed of the routes, 200,000 randomly selected OD pairs are searched and evaluated. For all the 200,000 OD pairs, the shortest-time route is different from the eco-route.

The environmental effectiveness of the eco-route is evaluated by the following indicators:

$$\alpha_{CO_2} = -(E_{CO_2} - T_{CO_2}) / T_{CO_2} \times 100\% \quad (13)$$

$$\alpha_{fuel} = -(E_{FC} - T_{FC}) / T_{FC} \times 100\% \quad (14)$$

$$\alpha_{time} = -(E_{TT} - T_{TT}) / T_{TT} \times 100\% \quad (15)$$

where α_{CO_2} , α_{fuel} , and α_{time} are the effectiveness of CO₂ emission reduction, fuel consumption reduction, and travel time reduction for the eco-route, respectively (the value of α_{time} is usually negative); E_{CO_2} , E_{FC} , and E_{TT} are the total amount of vehicle CO₂ emissions, fuel consumption, and travel time of the eco-route, respectively; and T_{CO_2} , T_{FC} , and T_{TT} are the total amount of vehicle CO₂ emissions, fuel consumption, and travel time of the shortest-time route respectively.

Table 3 Comparison between shortest-time route and eco-route for specific OD.

	Shortest-time route	Eco-route	%Differences
Travel Distance (km)	8.55	7.07	-17.3
Travel Time (min)	15.89	16.50	3.8
CO ₂ Emission (kg)	2.38	2.24	-5.9
Fuel Consumption (kg)	0.75	0.71	-5.3

Figure 6 shows the frequency distribution of effectiveness of CO₂ emission reduction for the four different distance ranges (0–5 km, 5–10 km, 10–15 km, and above 15 km). For the distance range of 0–5 km, the average effectiveness is 3.3%, and the most frequent effectiveness value is 1–2% with 16% frequency. For the distance range of 5–10 km, the frequency distribution of effectiveness is relatively dispersive. The most frequent effectiveness value is 8–9% with about 8% frequency, and the average effectiveness is 5.8%. For the distance range of 10–15 km, the effectiveness centralizes in the area from 3% to 9% with the average effectiveness 6.9%, while the effectiveness mainly centralizes in the range from 3% to 6% with the average effectiveness 5.6% for the distance range above 15 km. It can be concluded that the effectiveness of CO₂ emission reduction for 10–15 km travel distance range is the most significant on average.

The variation of the effectiveness under different average speed conditions is also analyzed. For all the examined OD pairs, the effectiveness of CO₂ emission reduction of the eco-route compared to the shortest-time route is calculated firstly using Eq. (13); then the results are grouped by the average speed of the shortest-time route in steps of 1 km/hr; and finally, the average effectiveness for each group is calculated, and the variation for average effectiveness of CO₂ emission reduction over the average speed is illustrated in Figure 7. The CO₂ emission reduction effectiveness increases and reaches the highest value of about 8% at the speed of about 48 km/hr, and begins to decrease with the increase of the average speed. This means at the average speed of about 48 km/hr, the eco-route reduces CO₂ emissions approximately 8% with the most significant environmental effectiveness.

Environmental Impact Assessment of Eco-Route Based on Traffic Status

The Travel Time Index (*TTI*) is defined by the Texas Transportation Institute (Schrank & Lomax, 2002) to describe the level of traffic congestion (Eq. (16)):

$$TTI = T_R / T_F \quad (16)$$

where T_F is the travel time in free flow conditions and T_R is the real travel time.

In this article, a travel speed index (*TSI*) is defined to evaluate the degree of the traffic congestion:

$$TSI = V_R / V_D \quad (17)$$

where V_R is the traveling speed and V_D is the design speed.

For any route, the average travel speed index (*ATSI*) is defined to evaluate the degree of traffic congestion of the whole route. The larger is the value of *ATSI*, the better is the traffic status:

$$ATSI = \sum_{i=1}^n TSI_i \times S_i / \sum_{i=1}^n S_i \quad (18)$$

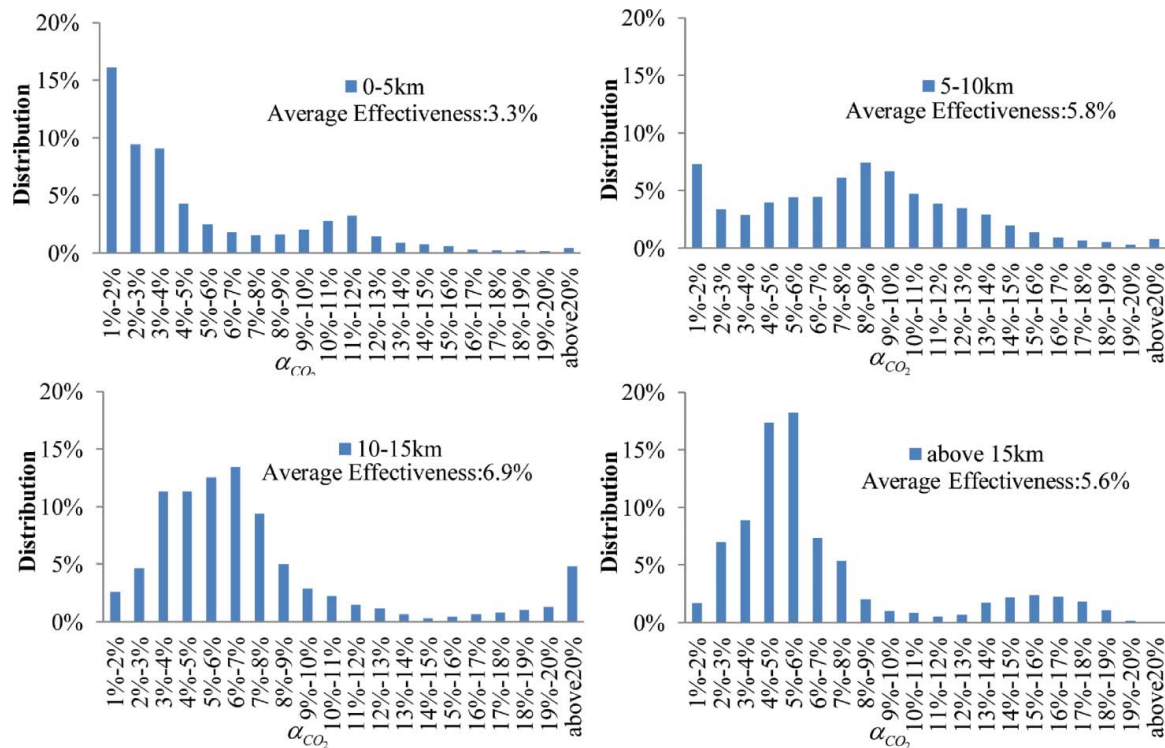


Figure 6 Distributions of effectiveness (CO_2) values under different distance ranges (color figure available online).

where TSI_i is the travel speed index for link i ; S_i is the length of link i ; and n is the number of links included in the route.

For the same 200,000 OD pairs mentioned earlier, route planning is executed for three types of vehicles. Figure 8 illustrates the variation in effectiveness (CO_2 , fuel, time) of the eco-route over the $ATSI$ for different vehicle types, and the influence of vehicle types on the effectiveness is obvious. The average effectiveness of different indicators is summarized according to the

$ATSI$ in steps of 0.1. The reductions of the CO_2 emission, fuel consumption, and the increment of travel time for light-duty vehicles are higher than the other vehicle types. The effectiveness values of CO_2 emission and fuel consumption reduction on average range from 5% to 8% for light-duty vehicles under all traffic conditions, and both reach the lowest value of about 5.5% when the $ATSI$ is about 1. In other words, the effectiveness values of the CO_2 emission and fuel consumption reduction are more obvious when the traffic condition is better or in heavy congestion. However, the change in travel time effectiveness value keeps relatively steady under all traffic conditions.

Table 4, Table 5, and Table 6 summarize the average values including CO_2 emission reduction effectiveness, fuel consumption reduction effectiveness, and travel time reduction effectiveness according to the categories of travel distance and travel traffic status ($ATSI$) for three types of vehicles respectively. The distance is divided into four ranges (0–5 km, 5–10 km, 10–15 km, and above 15 km), and the $ATSI$ that reflects the traffic status is divided into three parts (<0.9: heavy congestion; 0.9–1.1: moderate congestion; ≥ 1.1 : free flow). For all three types of vehicles, the average effectiveness values of CO_2 emission and fuel consumption reduction both reach the maximum under heavy congestion and 10–15 km travel distance conditions.

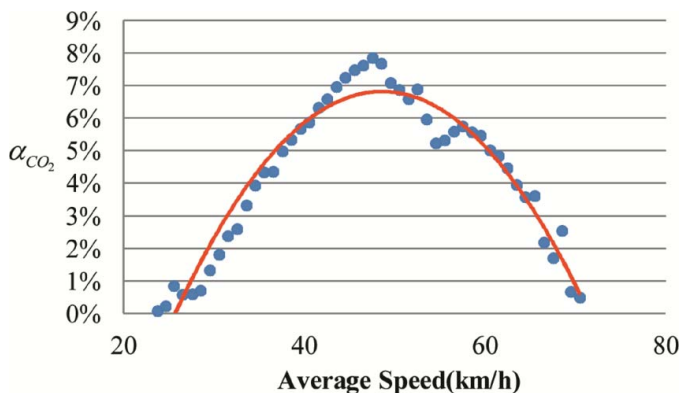


Figure 7 Effectiveness of CO_2 emission reduction under different average speed (color figure available online).

Table 4 Average effectiveness values of light-duty vehicles.

Distance	ATSI								
	<0.9 (Heavy congestion)			0.9–1.1 (Moderate congestion)			≥1.1 (Free flow)		
	CO ₂	Fuel	Time	CO ₂	Fuel	Time	CO ₂	Fuel	Time
0–5 km	3.3%	3.3%	–4.9%	3.2%	3.2%	–4.7%	4.0%	4.0%	–4.7%
5–10 km	5.8%	5.8%	–8.0%	5.7%	5.7%	–7.9%	6.0%	6.0%	–8.0%
10–15 km	7.3%	7.4%	–8.1%	6.1%	6.2%	–8.3%	6.0%	6.0%	–8.1%
Above 15 km	5.8%	5.9%	–6.4%	4.9%	4.9%	–6.5%	5.7%	5.7%	–7.0%

Bold-face type: Maximum CO₂ emission and fuel consumption reduction for light-duty vehicles.

Table 5 Average effectiveness values of mid-duty vehicles.

Distance	ATSI								
	<0.9 (Heavy congestion)			0.9–1.1 (Moderate congestion)			≥1.1 (Free flow)		
	CO ₂	Fuel	Time	CO ₂	Fuel	Time	CO ₂	Fuel	Time
0–5 km	2.7%	2.8%	–3.1%	2.6%	2.6%	–3.1%	3.0%	3.0%	–3.5%
5–10 km	3.7%	3.8%	–5.9%	3.8%	3.8%	–5.7%	3.9%	4.0%	–5.8%
10–15 km	4.5%	4.5%	–5.2%	3.6%	3.7%	–4.8%	3.6%	3.6%	–4.7%
Above 15 km	3.4%	3.4%	–4.2%	2.7%	2.7%	–3.7%	3.3%	3.3%	–4.1%

Bold-face type: Maximum CO₂ emission and fuel consumption reduction for mid-duty vehicles.

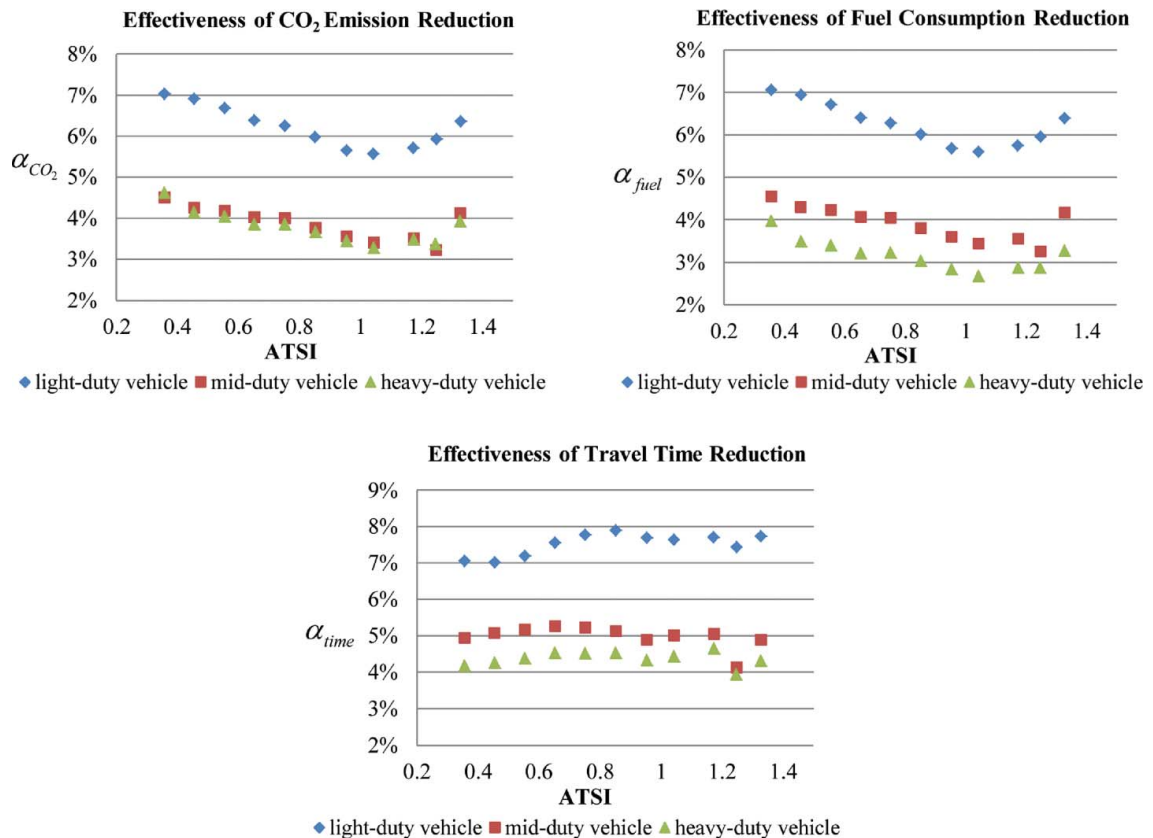
**Figure 8** Effectiveness under different traffic status (ATSI) of different vehicle types (color figure available online).

Table 6 Average effectiveness values of heavy-duty vehicles.

Distance	ATSI								
	<0.9 (Heavy congestion)			0.9–1.1 (Moderate congestion)			≥1.1 (Free flow)		
	CO ₂	Fuel	Time	CO ₂	Fuel	Time	CO ₂	Fuel	Time
0–5 km	2.6%	2.2%	–2.7%	2.4%	2.1%	–2.6%	2.9%	2.5%	–3.4%
5–10 km	3.2%	2.5%	–4.8%	3.3%	2.6%	–4.8%	3.5%	2.8%	–5.0%
10–15 km	4.7%	4.0%	–4.7%	3.8%	3.2%	–4.5%	3.7%	3.1%	–4.4%
Above 15 km	3.3%	2.8%	–3.7%	2.8%	2.3%	–3.4%	3.5%	3.0%	–3.9%

Bold-face type: Maximum CO₂ emission and fuel consumption reduction for heavy-duty vehicles.

CONCLUSIONS

In this study, mesoscopic vehicle emissions models and fuel consumption models for light-duty gasoline vehicles, mid-duty diesel vehicles, and heavy-duty diesel vehicles are established based on the PEMS data collected from the road network in Beijing. Through analyzing the VSP distribution for each travel speed level, the influence of the vehicle speed, acceleration, and other driving conditions on the vehicle emissions is fully analyzed, and the vehicle emission and fuel consumption models are established as functions of average travel speed. This method not only guarantees the accuracy of emission and fuel consumption models, but also makes it possible to estimate the emissions and fuel consumption based on most current traffic information systems, in which only the link-based travel time is available. Based on the environment-oriented link cost function and improved Dijkstra route planning approach, an eco-route planning algorithm is proposed. The effectiveness of CO₂ emission and fuel consumption reduction of the eco-route for all vehicle types is verified based on the results of the case study. The average effectiveness values of CO₂ emission reduction and fuel consumption reduction are both maximum under heavy congestion and 10–15 km travel distance conditions for all three types of vehicles. When average speed is about 48 km/hr, the greatest effectiveness reduction is achieved and about 8% amount of CO₂ emissions can be reduced for light-duty vehicles. It is very much expected that the proposed eco-route planning algorithm can contribute to the development of a low-carbon transportation system in China.

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